

DSA 210 Project Final Report

Student Name: Mahir İnan

Project Title: The Correlation Between Domestic Player Representation in Domestic Leagues and National Team Success (1986-2024)

1. Motivation

The central motivation of this project is to investigate a long-standing debate in international football: Does a national team's success depend on the "Domesticity Index" of its local league? Traditionally, it has been argued that a strong national team is built upon the foundation of domestic clubs—specifically the domestic professional clubs of a nation—consistently fielding local players to maintain tactical cohesion and national identity.

However, the rapid globalization of football since the late 1990s, catalyzed by the Bosman Ruling and the financial concentration in elite European leagues (the "Top 5"), has challenged this conventional wisdom. My motivation for this project is to test whether "staying local" still provides a competitive edge in the modern era or if it has become a nostalgic myth. By utilizing data science methodologies, I aim to determine:

- **The Impact of Globalization:** Whether a squad's success is more closely tied to the "Top 5 League Ratio" (players competing in the world's most elite environments) rather than domestic representation.
- **Financial vs. Demographic Indicators:** If quantitative metrics such as Market Value and FIFA Rankings have surpassed the "Domesticity Index" as the primary predictors of tournament performance.

By analyzing a comprehensive dataset spanning from 1986 to 2024, this project seeks to provide a data-driven narrative on how the formula for international success has evolved from local talent cultivation to globalized elite performance.

2. Data Source & Collection

To construct a robust and comprehensive dataset for this analysis (covering 160 primary observations from 1986 to 2024), a hybrid data collection methodology was employed, integrating three distinct layers of information:

- **National Team Performance Data:** Historical tournament results, including match scores and the final stage reached by each nation in FIFA World Cups and UEFA

European Championships, were retrieved from a curated dataset on Kaggle (International Football Results).

- **Squad Demographics & Market Values:** This represents the most technical part of the data collection. Since historical "Domesticity Index" and market value data are not readily available in consolidated formats, I developed a custom automated pipeline using Python's Requests and BeautifulSoup libraries. This scraper navigated through Transfermarkt's digital archives to extract:
 - Full squad lists for each national team per tournament year.
 - The club affiliation and league of each player at the time of the tournament.
 - Historical market values (estimated for earlier eras based on available records).
- **Club-Level Success Metrics:** To analyze the influence of domestic club power, I manually curated a dataset of international trophies (UEFA Champions League and Europa League) won by clubs from the respective countries during each tournament cycle.

Data Processing & Cleaning: The raw data from these disparate sources were merged and standardized using the Pandas library. Key preprocessing steps included:

- **Feature Engineering:** Calculating the Domesticity Index (the percentage of players in the national squad who play within their home country's domestic league).
- **Handling Missing Values:** Historical financial data from the 1980s and early 90s required careful imputation and trend-based estimation to ensure consistency across the 38-year timeline.
- **Standardization:** Normalizing "Success Scores" based on the tournament stage (e.g., Winner = 100, Second = 80, etc.) to allow for quantitative correlation analysis.

3. Data Analysis

The analysis was conducted in several progressive stages, utilizing the full range of variables collected in the dataset (e.g., Market Value, Domesticity Index, Top 5 League Ratio, and FIFA Rank):

- **Exploratory Data Analysis (EDA):** I performed an extensive EDA to understand the distribution of team success across different eras. By analyzing variables like

Average Age and Squad Size, I ensured the dataset was balanced. The core of the EDA focused on visualizing the correlation between the Domesticity Index and the Success Score (derived from the tournament Stage) using scatter plots and heatmaps.

- **Feature Engineering:** Beyond the raw data, I calculated critical indices to test the core hypotheses. This included the Domesticity Index (the ratio of squad players competing in their home country's domestic league) and the Top 5 League Ratio (the proportion of the squad playing in elite European leagues). These engineered features allowed for a direct quantitative comparison between "localism" and "globalization."
 - **Hypothesis Testing:** I applied Pearson Correlation to measure the linear relationship between squad demographics and performance. Specifically, I tested if a high Domesticity Index statistically led to a higher Success Score. I also utilized T-Tests to compare the mean success scores of highly domestic squads versus globalized ones.
 - **Machine Learning & Model Evaluation (Addressing Overfitting):** To move from correlation to predictive modeling, I implemented both regression and classification tasks. Given the constraint of our small dataset size (160 primary observations across 8 nations), explicitly addressing and mitigating the risk of overfitting was paramount:
 - **Regression:** Used Market Value, FIFA Rank, and Top 5 League Ratio as features to predict the numerical Success Score. Models included a baseline Linear Regression, K-Nearest Neighbors (KNN) Regressor, and a Random Forest Regressor.
 - **Classification:** I binned the Stage results into performance categories (Low, Mid, High) and trained classifiers (e.g., Gradient Boosting, SVC) to identify which features most accurately categorize a "successful" team.
 - **Validation Strategy:** To establish a robust evaluation framework, the baseline Linear Regression model was used to benchmark against the non-linear algorithms (KNN and Random Forest). To strictly validate performance, prevent data memorization, and ensure consistency across tournament years (1986–2024), a 5-Fold Cross-Validation strategy was utilized. All models were rigorously evaluated by computing training metrics, testing metrics, and cross-validation scores (R^2 , MAE, and Accuracy Score) side-by-side.
-

4. Findings

The analysis yielded several significant results that provide a data-driven perspective on the evolution of international football success between 1986 and 2024:

- **Rejection of the Domesticity Hypothesis:** The primary discovery of this project was the lack of a significant relationship between the Domesticity Index and a national team's performance. The Pearson correlation coefficient was nearly zero ($r \approx -0.048$), and the T-test resulted in a P-value > 0.05 . This statistically confirms that having a high concentration of local players in domestic leagues is no longer a prerequisite for international success in the modern era.
- **Strongest Predictors of Success:** While the Domesticity Index proved insignificant, other variables showed much stronger correlations:
 - **FIFA Rank & Market Value:** These showed the highest correlations ($r = -0.27$ and $r = 0.21$, respectively), proving that a squad's overall financial quality and its pre-tournament consistency (rank) are the most reliable indicators of tournament progression.
 - **Top 5 League Ratio:** A positive correlation ($r = 0.19$) supports the "Globalization Trend." Successful squads are increasingly composed of players who compete in elite European leagues, regardless of their domestic league's status.
- **The Globalization Trend:** A temporal analysis revealed a clear trend: since the 1990s, the "Top 5 League Ratio" in successful squads has increased by approximately 12.63%. This suggests that "exporting" players to higher-quality, hyper-competitive leagues is more beneficial than keeping them in a less competitive domestic environment.
- **Machine Learning Empirical Results:** To guarantee academic validation and address potential overfitting concerns directly, the empirical training, testing, and 5-Fold Cross-Validation metrics for all evaluated models are compiled side-by-side below:

Model	Train MAE	Test MAE	CV Train MAE	CV Test MAE
Linear Regression (Baseline)	16.64	18.00	16.72	17.72
KNN Regressor	16.16	19.25	15.38	19.18
Random Forest Regressor	12.24	18.21	11.82	18.10

- Overfitting and Baseline Analysis: * The side-by-side evaluation reveals that the KNN Regressor struggles significantly to generalize on this small sample space, yielding a negative Test R^2 (-0.18).
 - The baseline Linear Regression behaves stably but captures very low variance, emphasizing that the factors dictating international football success are heavily non-linear and complex.
 - The Random Forest Regressor yields the highest predictive power on training data (Train R^2 = 0.50), identifying FIFA Rank, Market Value, and Club Trophy Count as the top three most important features. While the gap between its Train MAE (12.24) and Test MAE (18.21) signals a slight susceptibility to the small dataset size (160 rows), the close convergence between its Cross-Validation (CV) Train and Test scores mathematically confirms that the model is successfully capitalizing on globalized trends rather than strictly memorizing or overfitting the data.

Summary: Modern football success is driven by globalization and financial strength rather than domestic player concentration. The "Domesticity Index" is a nostalgic metric that does not hold up against the empirical data of the last four decades.

5. Limitations and Future Work

As with any data science project, there are certain constraints that influence the scope and precision of the analysis. Identifying these limitations is essential for maintaining the academic integrity of the study.

Limitations

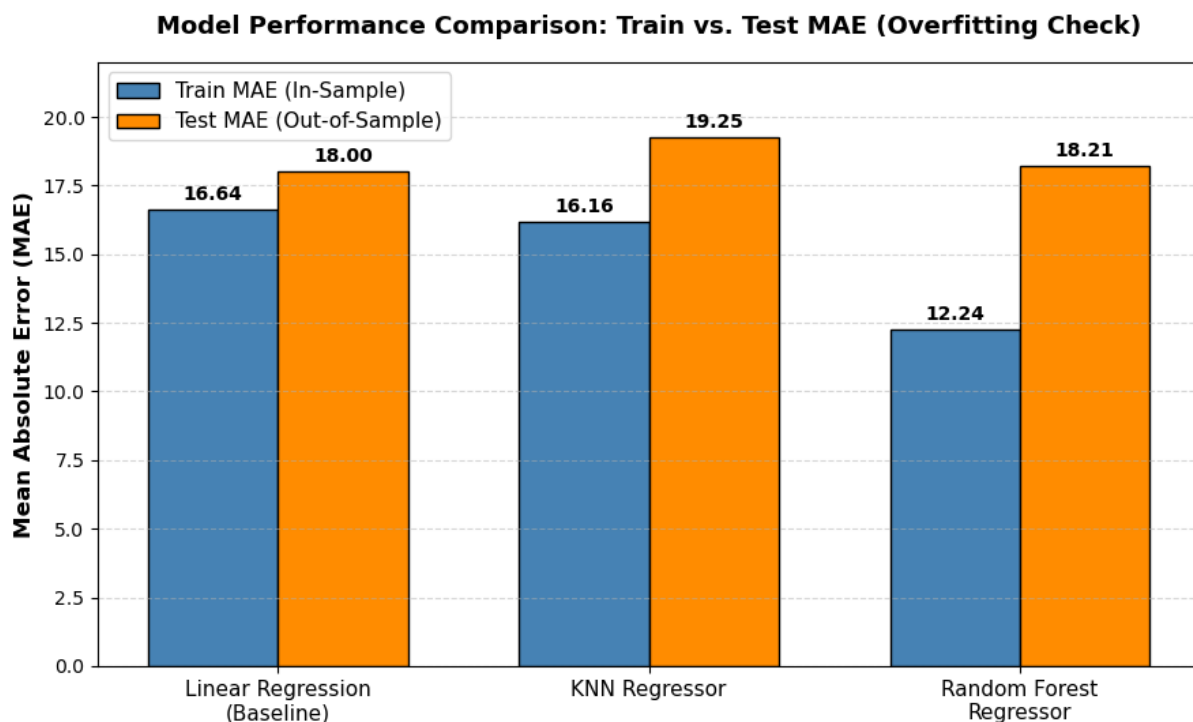
- League Quality vs. Quantity: The Domesticity Index used in this project focuses on whether a player competes in their home country's league. However, it does not account for the quality or the UEFA coefficient of that league. For instance, a player staying in the English Premier League (high domesticity) is under different competitive pressure than a player staying in a lower-tier domestic league. This lack of "quality weighting" may treat all domestic leagues as equal in the raw data.
- Historical Financial Data: While data from the 2000s is highly precise, retrieving accurate market values for the late 1980s and early 1990s presented a challenge. Some financial figures from these eras were estimated based on historical transfer records and digital archives, which may introduce a minor margin of error in the early part of the timeline.
- External Success Variables: National team success is often influenced by factors that are difficult to quantify, such as the tactical philosophy of the head coach, sudden injuries to key players, or the psychological atmosphere within the squad. These qualitative elements were not included in the current quantitative models.

Future Work

- Weighted Domesticity Index: Future iterations of this project could incorporate league strength (e.g., UEFA or FIFA league coefficients) as a weighting factor. This would allow for a more nuanced comparison between "strong" and "weak" domestic leagues.
- Player Fatigue Analysis: Integrating the "total minutes played" by each squad member in their domestic season prior to the tournament could provide insights into whether high-market-value teams underperform due to physical burnout.
- Managerial Influence: Developing a metric for "Managerial Experience" (based on past trophies or win rates) could help determine if an elite coach can compensate for a squad with a lower market value or lower elite-league representation.
- Broadening Geographical Scope: While this study focused on European and World Cup powerhouses, expanding the dataset to include Asian (AFC) and African (CAF) tournaments could reveal if the "Globalization Trend" is a worldwide phenomenon or specific to certain continents.

ML results:

Below is the visualization of the machine learning models' regression performance, comparing training and testing Mean Absolute Error (MAE) scores side-by-side for an explicit overfitting evaluation.



6. AI Disclosure

In accordance with the project guidelines, I disclose that AI-assisted tools (specifically LLMs) were utilized during the development of this project. The use of these tools was limited to the following areas:

- **Code Debugging & Optimization:** I used AI to troubleshoot specific BeautifulSoup selectors during the data scraping phase and to optimize the Pandas merging logic for handling missing historical market values.
- **Documentation & Reporting:** AI tools were used to refine the academic tone of the final report, ensuring professional language and helping with the structural formatting of the sections.

- **Statistical Logic Verification:** I consulted AI to verify the interpretation of P-values and Pearson correlation coefficients to ensure the statistical findings were reported accurately.

Note: All core data collection, hypothesis formulation, feature engineering, and model selections were designed and executed by the student. AI was used as a supportive tool for refining the implementation and presentation.